# Wenyuan Shi

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## Education

M.S. in Data Science | New York University Center for Data Science | New York, NY September 2026 - Present

- Courses taken (or to be taken) in: Methods in Quantitiative Biology, Healthcare Data Management, Machine Learning, Big Data

B.A. in Mathematics with Minor in Linguistics | Swarthmore College | Swarthmore, PA August 2022 - May 2026

- Cumulative GPA: 3.98/4.0; inducted into the Phi Beta Kappa honor society.
- Courses taken in: linear algebra, abstract algebra, multivariable calculus, real analysis, differential equations, mathematical modelling, probability, discrete mathematics, data structures & algorithms, computer systems

## Experience

Turing Lab Intern | Tencent June 2026 - August 2026

- Used **Python** to create new models to **analyse time series data** of the performance of users' devices, in order to help find areas where user experience can be improved. Increased $F_1$ score to greater than 80%. Results sent to dev team to help them find ways to reduce FPS stuttering, prevent crashing, etc.
- Used Python libraries including `pandas`, `statsmodels`, `sklearn`, and `matplotlib`.
- Practiced using **AI coding tools**.

Research Intern | JLSemi December 2025 - January 2026

- Used **Python** to analyse miscorrections in **error correcting codes**, specifically Reed-Solomon codes and Low-Density Parity-Check codes, and studied how the rate of miscorrections in RS codes changes given different parameters.

Undergraduate researcher | Department of Mathematics | University of Chicago June 2025 - August 2025

- Researched **algebraic topology** under the guidance of Professors Peter May (University of Chicago) and Carmen Rovi (Loyola University Chicago) and wrote paper on the relationship between Frobenius algebras and cohomology rings with Poincaré duality

Undergraduate researcher | Department of Mathematics and Statistics | Swarthmore College June 2024 - August 2024

- Worked with Professor Ian Whitehead and three other students to disprove the **Local-Global Conjecture in integral circle packings** of the octahedral, cubic, square, and triangular types.
- Wrote **C++** programs using **multithreading** to sort through circle packings and find counterexamples which disprove the conjecture. Also used **SageMath** with **Python** for helper scripts.
- Gave a **paper presentation** at the Joint Mathematics Meetings conference in 2025.
- Paper published in *Journal of Number Theory*; the DOI is `10.1016/j.jnt.2026.03.006`.

Undergraduate researcher | Department of Mathematics and Statistics | Swarthmore College June 2023 - August 2023

- Researched the structure of **hyperbolic space** with Professor Eduard Einstein by using **MATLAB** code to simulate random walks on a hyperbolic space with the Poincaré half-plane model and analysing the resulting data using regression analysis.

## Projects

Term Project – MATH 056 Modeling | Swarthmore College February 2026 - May 2026

- Created a model to analyse the spread of the invasive **emerald ash borer** (Agrilus planipennis) and its effects on Pennsylvanian ash tree populations and analysed the impact of various management techniques **Python** simulations of the model.
- Used Python libraries including `pandas`, `geopandas`, and `matplotlib`

Term Project – LING 073 Computational Linguistics | Swarthmore College February 2025 - May 2025

- Created an automatic interlinear gloss generator using **Apertium** and **Python**.

## Publications

- Hanqi Shi, Wenyuan Shi, Ian Whitehead, Ham Williams-Tracy, Jeffrey Zhirui Zhang, The Local-Global Conjecture is false for generalized circle packings, Journal of Number Theory, Volume 288, 2026, Pages 103-157, ISSN 0022-314X, `https://doi.org/10.1016/j.jnt.2026.03.006`.

## Leadership and Activities

Copresident | PsiPhi Club | Swarthmore College 2025

- Organised various events for the Swarthmore community, most notably the beloved Pterodactyl Hunt; the club also manages its own library with thousands of scifi and fantasy books

## Other Skills and Interests

- Programming & scripting languages: proficient with **Python**, **C++**, and **PostgreSQL**; moderately proficient with **C**, **MATLAB**, **HTML**, and **CSS**; some experience with **Java** and **C#** (and **x86 Assembly**, which I only used in a class)
- Experienced with using word processing, spreadsheets, presentations, and more in **Microsoft Office Suite** and **Google Workspace**
- Editing or creating images with **Photoshop**, **Krita**, or **GIMP**
- My Github: `github.com/Potato2357`